Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-6(P2) | Date: 16-04-26
Time: 3 Hrs | 2024_P2 | Max. Marks: 180

KEY SHEET
MATHS

1	**A**	2	**B**	3	**B**	4	**D**	5	**BCD**
6	**ACD**	7	**AD**	8	**8**	9	**31**	10	**65**
11	**60**	12	**58**	13	**3**	14	**132**	15	**282**
16	**13**	17	**35**						

PHYSICS

18	**A**	19	**B**	20	**D**	21	**D**	22	**BCD**
23	**AC**	24	**ABCD**	25	**1**	26	**9**	27	**6**
28	**3**	29	**6**	30	**4**	31	**3.46 TO 3.47**	32	**5.35 TO 5.36**
33	**0.75**	34	**0.50**						

CHEMISTRY

35	**D**	36	**C**	37	**B**	38	**C**	39	**AB**
40	**ABCD**	41	**AC**	42	**232**	43	**960**	44	**5**
45	**5**	46	**7**	47	**6**	48	**33**	49	**247**
50	**1308 TO 1311**	51	**1643 TO 1647**						

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SOLUTIONS
MATHS

1. $L=\int_{-\pi/2}^{\pi/2}\frac{x^4\cos x}{x^2}dx=\int_{-\pi/2}^{\pi/2}x^2\cos x dx=2\int_0^{\pi/2}x^2\cos x dx$

$=2\left(\left(x^2\right)(\sin x)-(2x)(-\cos x)+2(-\sin x)\right)_0^{\pi/2}=2\left(\frac{\pi^2}{4}-2\right)=\frac{\pi^2}{2}-4$

2. $z_1^2=z_2z_3,\ z_2^2=z_1z_3\Rightarrow z_3^2=z_1z_2$

$\therefore\sum z_1^2=\sum z_1z_2\Rightarrow\triangle ABC$ is equilateral

Given side $=3\Rightarrow$ Area $=\frac{\sqrt{3}}{4}(3)^2=\frac{9\sqrt{3}}{4}$

3. $S=\sum_{n=1}^{\infty}\tan^{-1}\left(\frac{(n+1)(n+2)-n(n+1)}{1+n(n+1)(n+1)(n+2)}\right)=\tan^{-1}\left((n+1)(n+2)\right)=\tan^{-1}\left(n(n+1)\right)$

$\sum_{n=1}^{\infty}\left(\phi(n)-\phi(n-1)\right)$

$=\frac{\pi}{2}-\tan^{-1}2=\cot^{-1}2=\tan^{-1}\frac{1}{2}=\sin^{-1}\left(\frac{\sqrt{5}}{2}\right)$

4. $\sqrt{x^2-(r-r_1)}=r+r_1\Rightarrow x^2=4rr_1$

$\sqrt{p^2+(r-r_2)^2}=r-r_2\Rightarrow p^2=4rr_2$

$\sqrt{(x-p)^2+\left(2r-(r_1+r_2)\right)^2}=r_1+r_2$

$\Rightarrow 4r^2=2px\Rightarrow 2x^2=px\Rightarrow 2r^2=\sqrt{4rr_1}\sqrt{4rr_2}\Rightarrow r=2\sqrt{r_1r_2}$

$\therefore r=2\sqrt{5\sqrt{2}+10\sqrt{2}}=2(10)=20$

5. Tgt at Q :

$x-2y+4=0\Rightarrow Q=(4,4)\Rightarrow\alpha=4$

$\therefore P(-1,4)$ lies on directrix (PAB right angled triangle)

$\triangle PAB$ orthocenter is $(-1,4)\Rightarrow$ distance from origin $\sqrt{17}$

$QM=3\Rightarrow TN=2$ also correct.

6. $GI=\frac{(2+\sqrt{3})^{2026}-\left(1+\sqrt{8}\right)2026}{2026}\Rightarrow a=2,b=3,c=8,d=2025$

$=5^2.3^4$

8. $(a+x)(b+x)(c+x)(d+x)-\left(x^2+x+1\right)=(x-1)(x-2)(x-3)(x-4)$

$G.E=(31+24)(43+120)=(55)(163)=5'11'163'$

Number of divisors $=2\times2\times2=8$

9. $f(x)$ non - differential at $x=-2,-6$

$f(-|x|)$ non - differentiable at $n=\pm2,\pm6$

Also discontinues at $x=0$ (non - diff)

$\therefore p=1,\ q=5,\Rightarrow G.E=1^2+(1)(5)+5^2=31$

10. $P(\bar{o}),Q(\bar{q}),R(\bar{r}),S(\bar{s})$

$\therefore\left|\bar{q}\right|=6,\left|\bar{r}\right|=4,\left|\bar{s}\right|=3,\left|\bar{r}-\bar{q}\right|=4,\left|\bar{x}-q\right|=5,\left|\bar{s}-\bar{r}\right|=2$

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$$A=\left(\frac{\vec{q}}{2}\right), B=\frac{\left(\vec{r}+\vec{s}\right)}{2} \Rightarrow A.B=\frac{1}{2}\left|\vec{r}+\vec{s}-\vec{q}\right| \Rightarrow AB=\sqrt{\frac{13}{2}}=k$$

$$\Rightarrow 10k^2=65$$

11. $\left.\begin{matrix} m &= 50-13=37, \\ n &= 50-27=23 \end{matrix}\right\} \Rightarrow m+n=37+23=60$

12. Consider O also

$$GE=(p+\sqrt{q})^2+(p-\sqrt{q})^2=2\left(p^2+q\right)=2\frac{\sum x_i^2}{n}$$

$$=2\cdot\left(0^2+1^2+4^2+9^2+6^2+5^2+6^2+4^2+1^2\right)+()+()+\ldots\ldots+0^2+1^2+4^2+9^2+6^2+5^2$$

$$=\frac{2}{2025}\left[(202)(293)+159\right]=58.61=k \Rightarrow [k]=58$$

13. $f(N)=1-\frac{1}{\log_2(N+1)}$

$$G.E=\frac{1}{\left(1-\frac{1}{3}\right)\left(1-\frac{1}{6}\right).....\left(1-\frac{1}{14}\right)}=\frac{1}{\left(\frac{4}{14}\right)}=\frac{7}{2}$$

$I.P=3$

14,15: $|z_1-z_2|=|z_2-z_3|=|z_3-z_1|=2\sqrt{3}$

$$\left(z_1+z_2+z_3\right)\left(\vec{z_1}+\vec{z_2}+\vec{z_3}\right)=441$$

$$\Rightarrow \sum|z_1|^2+\sum\left(z_1\vec{z_2}-z_2\vec{z_1}\right)=441 \Rightarrow 27+q$$

$$\Rightarrow p+q=414\#(1)$$

Also $\sum|z_1-z_2|^2=3\times(12)=36 \Rightarrow 2\sum|z_1|^2-p=36 \Rightarrow 2(27)+2.q-p=36$

$$\Rightarrow 2q-p=-18\#(2)$$

Solving $3q=396 \Rightarrow q=132, p=282$

16,17. $\int_0^{\ln 2} e^{-2x}f''(x)\,dx=e^{-2x}f'(x)\Big|_0^{\ln 2}-\int_0^{\ln 2}e^{-2x}(-2)f'(x)\,dx$

Also $\int_0^1(g(x))^2-2\left(2x^2\right)g(x)+\left(2x^2\right)^2-4x^4\,dx=-\frac{9}{5}$

$$\Rightarrow \int_0^1\left(g(x)-2x^2\right)^2dx=0 \Rightarrow g(x)=2x^2$$

$$\therefore A.M=\frac{\left(\frac{1}{2}+\frac{1}{2^3}+\frac{1}{2^5}+.....+\frac{1}{2^{19}}\right)}{10}=\frac{\frac{1}{2}\left(1-\left(\frac{1}{2^2}\right)^{10}\right)}{\left(1-\frac{1}{2^2}\right)}$$

$$=\frac{1}{15}\left(1-\frac{1}{2^{20}}\right) m=15, n=20$$

$m+n=35$

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PHYSICS

19. The wavelength λ_{min} emitted by the X -ray tube operating at a voltage V is given by $eV = \frac{hc}{\lambda_{min}}$

Kinetic energy of the electron $= \frac{1}{2}mv^2 = \frac{1}{2}\left(\frac{m^2v^2}{m}\right) = \frac{h^2}{2m\lambda_1^2}$

Now the wavelength λ of an electron moving with the velocity v is given by

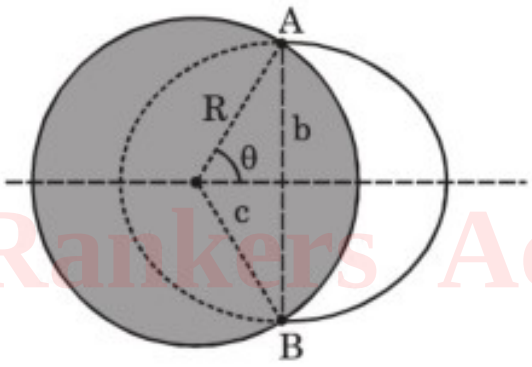
$\lambda = \frac{h}{mv} \Rightarrow \lambda = \frac{h}{\sqrt{2meV}}$

$\therefore$ K.E. $= 600eV$

19. $\frac{A_1}{t_1} = \frac{A_2}{t_2}$

$\frac{t_2}{t_1} = \frac{A_2}{t_1} = \frac{\frac{\pi ab}{2} + \frac{1}{2}(2b)(c)}{\pi ab}$

$\frac{t_2}{t_1} = \frac{1}{2} + \frac{c}{a\pi} = \frac{1}{2} + \frac{\cos\theta}{\pi}$



$= \frac{1}{2} + \frac{1}{a\pi} = \frac{1}{2}\left(1 + \frac{1}{\pi}\right)$

$= \frac{1}{2}\left(1 + \frac{1}{3}\right)$

$\frac{t_2}{t_1} = \frac{2}{3}$

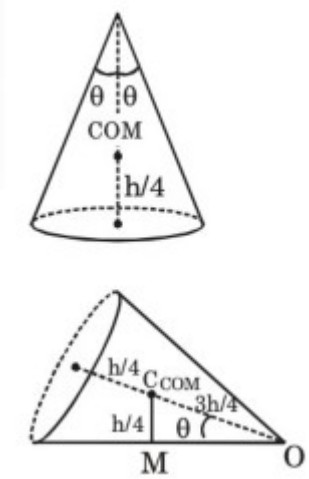
20. $d\vec{B} = \frac{\mu_0 I d\vec{l} \times \vec{r}}{4\pi r^2}; dB = \frac{\mu_0 I}{4\pi}\frac{r d\theta}{r^2}$

$dB = \frac{\mu_0 I}{4\pi}\frac{d\theta}{\left(a + \frac{b}{\pi}\theta\right)}; B = \int dB = \frac{\mu_0 I}{4\pi}\int_0^{\pi}\frac{d\theta}{a + \frac{b}{\pi}\theta}$

$= \frac{\mu_0 I}{4\pi}\frac{\pi}{b}\left[\ln\left(a + \frac{b}{\pi}\theta\right)\right]_0^{\pi} = \frac{\mu_0 I}{4\pi}\left[\ln(a+b) - \ln a\right] = \frac{\mu_0 I}{4b}\ln\left(1 + \frac{b}{a}\right)$

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21. Solid Cone



$$\sin\theta = \frac{h/4}{3h/4} = \frac{1}{3}$$

$$\therefore \tan\theta = \frac{1}{2\sqrt{2}} = \frac{R}{h}$$

$$\therefore \frac{h}{R} = 2\sqrt{2}$$

22. $f = f_0 \dfrac{C+v}{C-v}$

$$\therefore \lambda = \frac{C(C-v)}{f_0(C+v)}$$

Beats frequency $= \Delta f = f_0 \dfrac{C+v}{C-v} - f_0 = f_0\left(\dfrac{C+v-C+v}{C-v}\right) = \dfrac{2f_0 v}{C-v}$

23. If v be the velocity at mean position in two cars then $\frac{1}{2}mv^2 = \frac{1}{2}KA_1^2 = \frac{1}{2}KA_2^2$ and $T_1 = T_2 = 2\pi\sqrt{\dfrac{m}{K}}$

$A_1 = A_2$

24. According to gauss law $E4\pi r^2 = \dfrac{q_0 - Q_1}{\epsilon_0}$

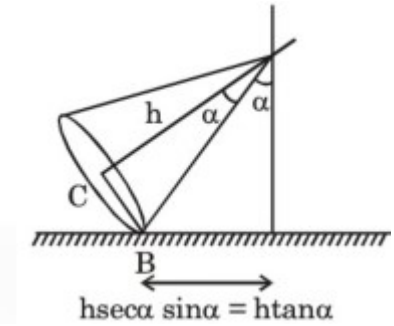
$$J = \frac{E}{\rho} = \frac{q_0 - Q_1}{4\pi\epsilon_0 \rho r^2}$$

$$I = JA = \frac{1}{\epsilon_0 \rho}(q_0 - Q)$$

$$I = \frac{dQ}{dt} \Rightarrow \frac{dQ}{q_0 - Q} = \frac{dt}{\epsilon_0 \rho} \Rightarrow \frac{Q}{q_0} = 1 - e^{-t/\epsilon_0 \rho}$$

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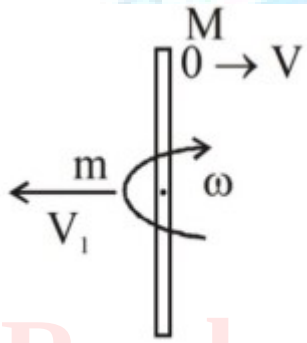
25.



$\omega_1 h\tan\alpha = \omega_2 R$

$\Rightarrow \frac{\omega_1}{\omega_2}\frac{R/h}{\tan\alpha} = 1$

26. i) $L_i = L_f \; '\frac{m\ell^2}{12}\omega = MV \times \frac{1}{2} \Rightarrow V = \frac{m\omega\ell}{6M}$

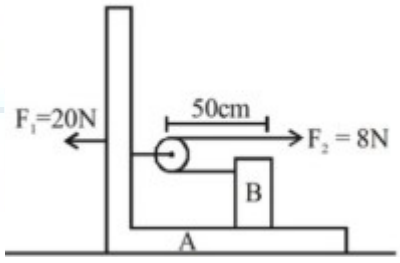


ii) $P_i = P_f; 0 = MV - mV_1 \Rightarrow V = \frac{mV_1}{M} \Rightarrow \frac{\omega\ell}{6} = V_1$

iii) $e-1 = \frac{V-(-V_1)}{\omega\frac{1}{2}-0} \Rightarrow V+V_1 = \frac{\omega\ell}{2}, \frac{m\omega\ell}{6M}+\frac{\omega\ell}{6} = \frac{\omega\ell}{2}; \left(\frac{m}{M}+1\right) = 3$

$\Rightarrow \frac{m}{M} = 2 \Rightarrow M = \frac{m}{2} = 9 \text{ kg}$

27.



$a_A = \frac{20-16}{1} = 4 \text{ m/s}^2$

$a_B = \frac{8}{1} = 8 \text{ m/s}^2$

$a_{B/A} = 4m/s^2$

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$$0.5 = \frac{1}{2} \times 4t^2$$

$$t = \frac{1}{2}\text{sec} = 0.5\text{sec}$$

28. As $v_2 = \dfrac{2m_1}{m_1 + m_2} v_1$

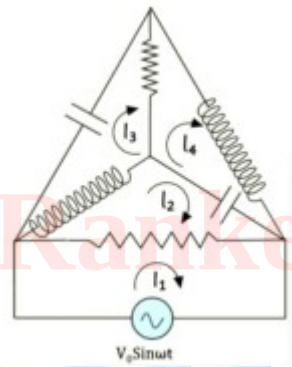
$$\therefore v_3 = \frac{2m_2}{m_1 + m_2} v_2 = \frac{4m_1 m_2 v}{(m_1 + m_2)(m_2 + m_3)}$$

For maximum $v_3, \dfrac{dv_3}{dm_2} = 0 \Rightarrow m_2 = \sqrt{m_1 m_3} = 3$ kg

29. $$\frac{(\sigma_2^2 - \sigma_1^2)}{2\epsilon_0}\left(1 - \frac{1}{K}\right) = \rho g h$$

$$\Rightarrow h = \frac{3}{4}\frac{\sigma^2}{\epsilon_0 \rho g}$$

30.



$$x_L = x_C = R\left(\omega = \sqrt{\frac{L}{C}}\right)$$

$$v_0 = (i - i_2)R = j(i_4 - i_3)R\left[j = \sqrt{-1}\right]$$

$$-ji_3 + ji_4 - j(i_4 - i_3) + j(i_3 - i_2) = 0$$

$$\Rightarrow i_2 = i_3$$

$$i_4 - i_3 + ji_4 - j(i_4 - i_2) = 0$$

$$i_4 - i_2 + ji_2 = 0$$

Upon solving $\rightarrow i_1 = \dfrac{2V_0}{R} = 1.6$ A

31. From energy conservation- $mCT_1 + mCT_2 = 2m\left(CT + \dfrac{1}{2}v^2\right) v = \sqrt{C(T_1 + T_2 - 2T)}$

32. The maximum speed occurs when the process is internally reversible, meaning there is no entropy generation.

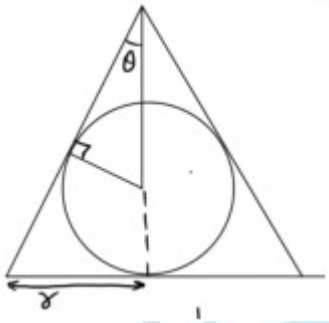
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$$mC\ln\left(\frac{T}{T_1}\right)+mC\ln\left(\frac{T}{T_2}\right)\geq 0$$

$$T=\sqrt{T_1T_2}$$

$$u_{\max}=\sqrt{C\left(\sqrt{T_1}-\sqrt{T_2}\right)^2}=\left(\sqrt{T_1}-\sqrt{T_2}\right)\sqrt{C}$$

33.



$$\sin\theta=\frac{1}{2}$$

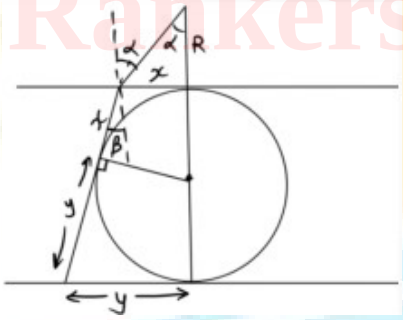
$$\Rightarrow\theta=30^\circ$$

$$\frac{r}{3R}=\tan\theta$$

$$\Rightarrow r=\sqrt{3}R$$

$$A=\pi r^2=3\pi R^2$$

34.



$$\sin\alpha=\sqrt{3}\sin\beta$$

$$(x+y)^2=(y-x)^2+4R^2$$

$$xy=R^2$$

$$\frac{R^2/y}{\sqrt{R^2+\frac{R^4}{y^2}}}=\frac{\sqrt{3}\left(y-\frac{R^2}{y}\right)}{\left(y+\frac{R^2}{y}\right)}$$

$$R\left(\frac{y^2+R^2}{y}\right)=\sqrt{3\left(y^2+R^2\right)}\left(\frac{y^2-R^2}{y}\right)$$

$$\Rightarrow y^2=2R^2$$

$$A_s=\pi y^2=2\pi R^2$$

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CHEMISTRY

35. NCERT Page No. 434

36. C.N, density and distance beetween closest octahedral voids (2 r) in same for both hcp and *fcc* unit cells.

37. Cu^{2+} gives Insoluble Cu_2I_2

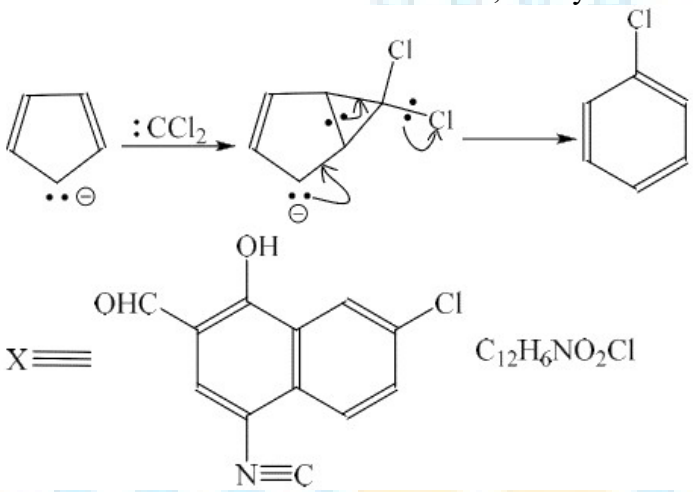
38. SnF_4, PbF_2, PbF_4 are ionic compounds

39. NCERT Page No. 453

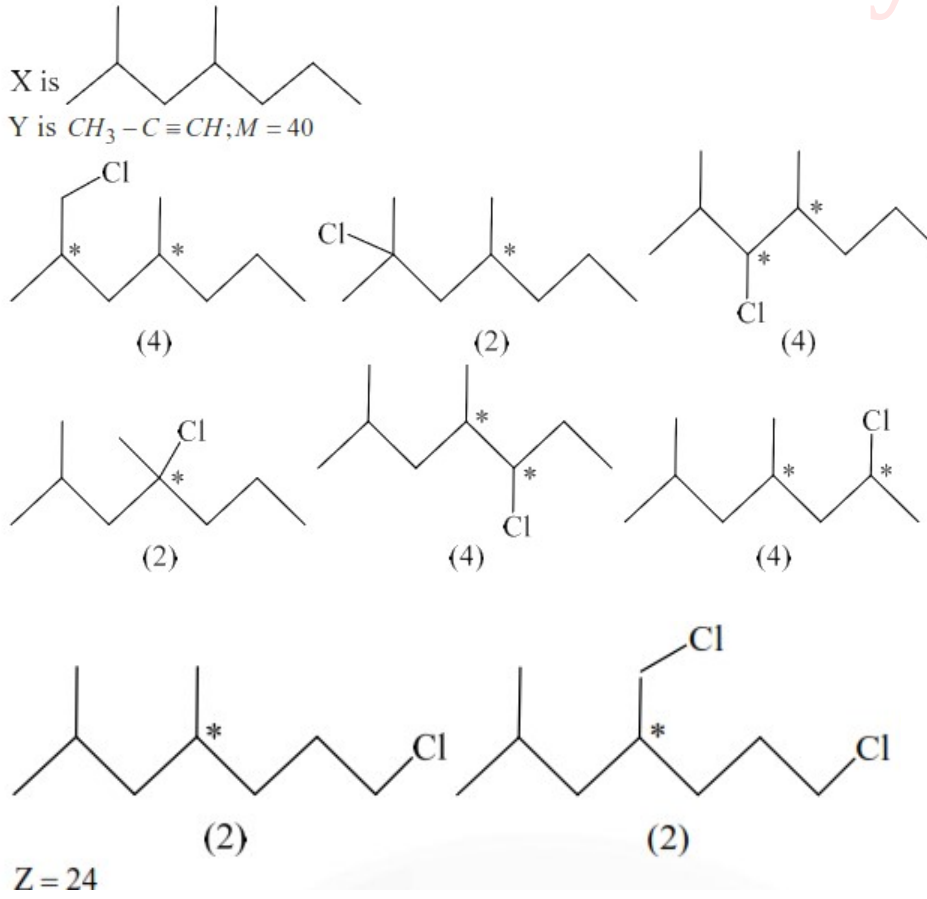
41. B is iodine, C is $S_4O_6^{2-}$

G is $K_2[PbI_4]$ and E is KI_3

42. It is the combination of Reimer - Tieman, Carbylamine reaction and the following reaction



43.



Z = 24

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44. I vapour:

$$^{Y}C_2 = \frac{X_c P_c^{\circ}}{X_c P_c^{\circ} + X_b P_b^{\circ}} = \frac{\frac{2}{5}\times 300}{\frac{2}{5}\times 300 + \frac{3}{5}\times 200} = 0.5$$

Last Bubble:

$$^{Y}C_2 = \frac{2}{5} = 0.4$$

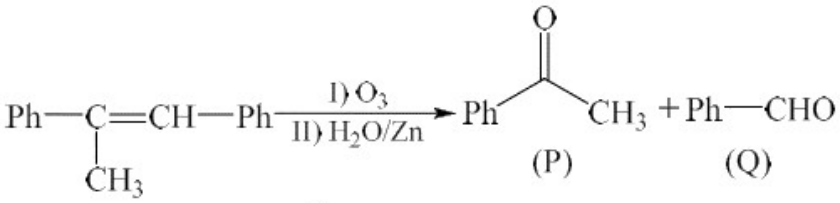
$$\frac{^{Y}C_1}{^{Y}C_2} = \frac{0.5}{0.4} = 1.25$$

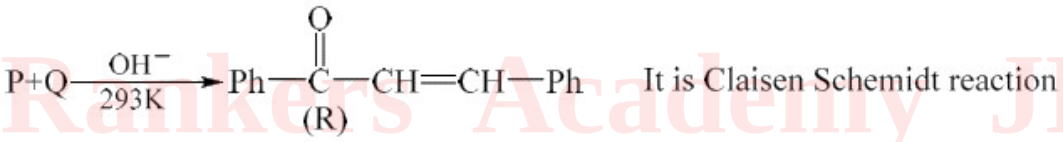
45. Cr is only reduced, ' O ' is both oxidized and reduced.

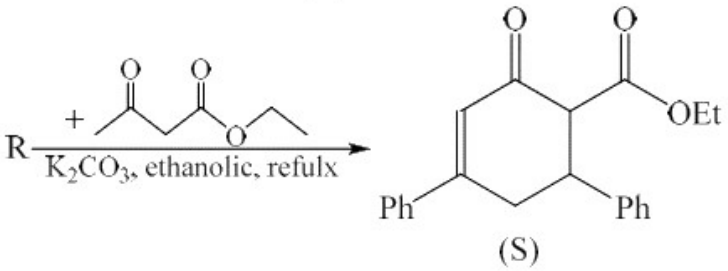
46. $\left(Cu\left(NH_3\right)_4\right)^{2+}$ number of axial electrons are equals to 3 where as $\left(Ni(CN)_4\right)^{4-}$ axial ions are equal to 4

47 $H_2O_2, PCl_3, SF_4, XeO_3, PH_4^+, \left[Al(OH)_4\right]^-$

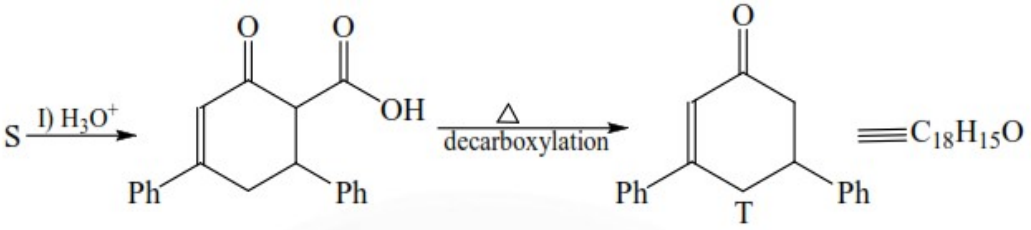
48.







It is Robinson's annulation



M.W.=247

D.U of R is 10, S is 12, T is 11

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50. For $2Fe + O_2 + 2H_2O \rightleftharpoons 2Fe^{2+} + 4OH^-; E^\circ = 0.84$ V

$$E_{cell} = 0.84 - \frac{0.06}{4}\log\frac{\left[Fe^{2+}\right]^2\left[OH^-\right]^4}{p_{O_2}}$$

$$= 0.84 - \frac{0.06}{4}\log\frac{\left[10^{-2}\right]^2\left[10^{-7}\right]^4}{0.2}$$

$$= 1.3095 \text{ V}$$

$$= 1309.5\text{mV}$$

51. $SO_2(aq) + H_2O(l) \rightleftharpoons H^+ + HSO_3^-$ $K = 1.6\times10^{-2}$

eql $0.1MCC$

$$\left[H^+\right]^2 = 0.1\times1.6\times10^{-2}$$

$$\Rightarrow \left[H^+\right] = 4\times10^{-2}$$

$$\Rightarrow \left[OH^-\right] = \frac{1}{4}\times10^{-12}$$

$$E_{cell} = 0.84 - \frac{0.06}{4}\log\frac{\left[Fe^{2+}\right]^2\left[OH^-\right]^4}{p_{O_2}}$$

$$= 1.6455 \text{ V} = 1645.5\text{mV}$$

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